

Reference book

- ① Software Engineering: A Practitioner's Approach — Roger S. Pressman (8th Semester)
- ② Ian Sommerville (9th Edition) — Software Engineering

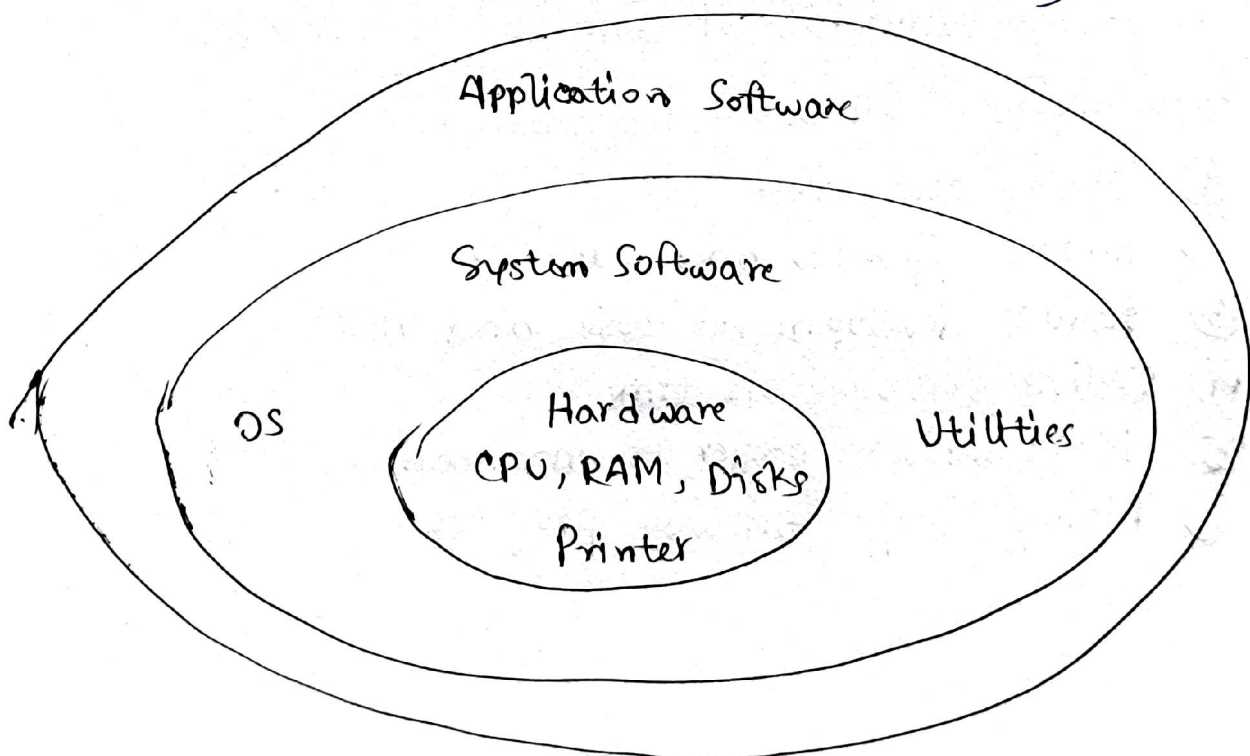
→ Software

Software = Programs + Related data + Documentation

- ① Instruction
- ② Data Structures
- ③ Descriptive Information

→ Software Application Domains

- ① System software (OS)
- ② Application software (Needed by the user)



→ Software Engineering

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software.

It is not only coding; it covers the whole software life cycle.

→ Software Process

A set of activities used to develop and maintain software. Main activities:

- ① What should we build? (Specification)
- ② (Development) // How do we build it?
- ③ (Validation) // check whether it meets requirements
- ④ (Evolution) // Modify as needs change.

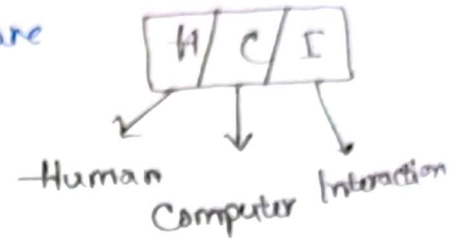
→ Fundamental Activities of Software Engineering

→ Importance of Software Engineering

- ① Control complexity
- ② Improve quality and reliability
- ③ Reduce development cost and risk
- ④ Deliver software on time
- ⑤ Make software easier to maintain
- ⑥ Manage large software projects.

➤ Essential Attributes of a Good Software

- ① Functionality
- ② Usability
- ③ Portability
- ④ Scalability
- ⑤ Flexibility
- ⑥ Security
- ⑦ Performance
- ⑧ Reliability



➤ Legacy Software

An older version of a program or application that is used by organization for various reasons.

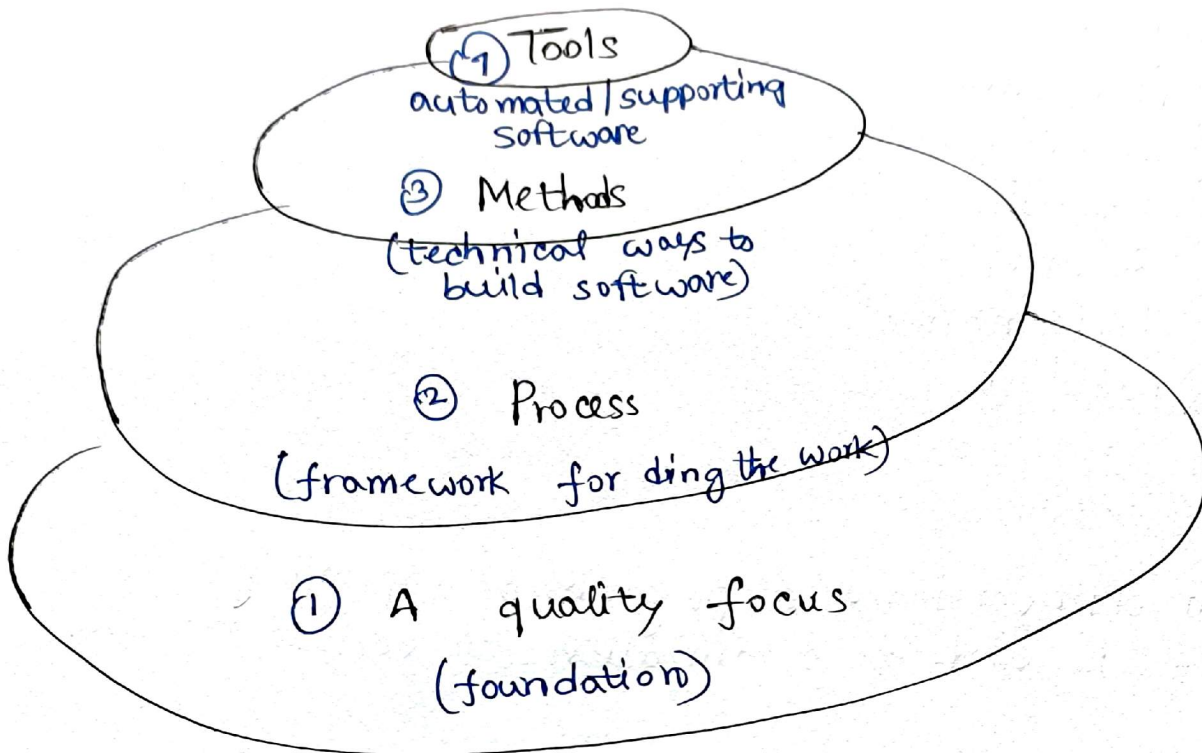
➤ Used by

➤ Software Engineering Ethics

- ① Confidentiality
- ② Competence (Don't over promise)
- ③ Intellectual property rights
- ④ ~ computer misuse

➤ FAQ About Software

→ Software Engineering Layers



→ Process Framework

- ① Framework Activities
- ② Umbrella Activities

→ Framework Activities

- ① Communication
- ② Planning
- ③ Modeling
 - Analysis of requirements
 - Design
- ④ Construction
 - Code generation
 - Testing
- ⑤ Deployment